

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0755

Burle 138 kV
November 20, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0755. The Interconnection Customer proposes a 222.5 MW natural gas combined cycle facility interconnecting at the Burle 138 kV substation in the Pennsylvania area. The request was received on March 06, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 268 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,378,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Grizzly to Lockeford 138 kV L
Contingency: Fayette to Mercury 345 kV L
Loading: 125.7% (Rating: 197 MVA)
DFAX: 7.6%
Pre-project loading: 117.2%
2. Facility: Leola to Sharyland 138 kV
Contingency: Sharyland to Bixby Hill 345 kV line
Loading: 124.8% (Rating: 222 MVA)
DFAX: 22.8%
Pre-project loading: 102.0%
3. Facility: Del Cerro to Grizzly 138 kV
Contingency: Mercury to Faette 345 kV
Loading: 122.4% (Rating: 214 MVA)
DFAX: 7.6%
Pre-project loading: 114.6%
4. Facility: Leola to Burle 138 kV
Contingency: Sharyland to Bixby Hill 345 kV L
Loading: 111.4% (Rating: 228 MVA)
DFAX: 45.1%
Pre-project loading: 67.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Grizzly to Lockeford 138 kV L	Rebuild line with higher-capacity conductor	\$958,000	26% (\$250,000)
Sharylad to Leola Substation 138 kV	Rebuild line with higher-capacity conductor	\$23,474,000	100% (\$23,474,000)
Del Cerro to Grizzly 138 kV	Rebuild line with higher-capacity conductor	\$3,473,000	26% (\$906,000)
Leola to Burle 138 kV	Rebuild line with higher-capacity conductor	\$3,748,000	100% (\$3,748,000)
	Total allocated to project		\$28,378,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.